DASR Supplement 3: Baseline Lineage, Formulation Parity, and Protocol Details

Anonymous

1 Common evaluation protocol

All methods receive the same frozen cohort, four evidence tiers, cumulative feature costs, false-positive and review prices, terminal full-evidence reference, and public audit vocabulary. The four data roles are Cascade Training, Policy Construction, Policy Selection, and Sealed Audit. Policy Construction supplies each method's complete frozen policy menu.

Each adaptive method creates its own finite, ordered policy menu on Policy Construction data. Policy Selection evaluates each frozen member at the common pointwise 0.045 tail screen and chooses the least-cost passing member. Exactly one selected policy from each reported method accesses Sealed Audit, where the common 0.05 boundary is applied. Candidate-menu size is retained as provenance. The four fixed-tier controls each contain one nonadaptive policy.

This design compares routing rules under the same operational problem. The common screen and audit supply formulation parity for the slice-tail estimand.

2 Registered-batch execution specification

A DASR occupation-measure policy may store more than one positive action share at a reached state. The reference executor realizes these shares within each registered batch by calibrated-score ordering, deterministic integer allocation, and stable tie-breaking. Batch identity, independent-unit grouping, ordering, rounding, and implementation version are therefore part of the execution specification. Pure rows reduce to ordinary casewise actions.

All reported full-batch replays match the stored selected decisions exactly. The selected-policy execution audit finds 8 pure selected policies and 19 policies with at least one mixed occupied row. Deterministic partition checks characterize the cohort-relative allocations of mixed rows. Governance edits receive complete-batch replay verification for zero off-address route changes, followed by the common fail-closed screen and audit.

3 Comparator lineage and adaptations

Paper label	Source principle	Registered adaptation
Fixed T_0–T_3	Nonadaptive operating controls	Every case stops at the named tier and receives the common terminal action rule. Each tier is reported separately.
Score-threshold cascade	Scalar confidence routing	Tier-specific calibrated-score thresholds create a construction-only menu over early termination and continuation.
Fast Yet Safe	Risk-controlled early exit [3]	The primary signed excess-miss formulation matches the paper's one-sided reference harm. A clipped positive-part formulation is evaluated separately as sensitivity. Both use the common 0.045/0.05 protocol and slice-CVaR evaluation.
MSRC	Sequential rejectors [5]	Stagewise reject decisions use exact downstream evidence-cost offsets, the shared four-action vocabulary, terminal reference, and common screen/audit.
Wang sequential LP	Joint sequential classifier LP [6]	The convex sequential program uses the registered evidence and action costs and emits a method-local menu over its regularization/control parameter.
Ringel accumulated-gap control	Conditional accumulated-gap early exit [4]	Tier-specific accumulated-gap thresholds are fit on Policy Construction and evaluated under the common economic and tail-risk specification.

The calibrated-score representation is the direct matched state-resolution ablation: it uses the DASR occupation-measure routing program restricted to public one-dimensional calibrated-score bins.

4 Structural-comparison lineage

The dense raw-prefix comparison uses direct estimates for every supported current-tier model prefix. Prefix-HS and path-HS apply the Hierarchical Shrinkage estimator of Agarwal et al. [1] to current-tier prefixes and cumulative histories. The main sensitivity includes the published shrinkage range and a separately labeled expanded range. All settings receive the same common screen and audit.

DASR begins with contract-induced calibrated decision regions, proposes balanced score refinements and complete model-native child/complement releases, screens them with the fixed-dual action-face ledger, and retains an admission only after an exact constrained LP re-solve changes construction routes and improves the objective. Its final frontier contains calibrated-score, independently constructed path-free, and path-bearing classes.

5 Audit functional

The public audit contains 40 predeclared slices: ten calibrated-score bins crossed with four stopping tiers. Within each slice, the harm event is a stopped automated negative that disagrees with the full-evidence reference on a positive case. Slice rates receive one-sided exact Clopper–Pearson upper bounds [2]; the reported statistic averages the worst 20% of stopped-negative mass. With total slice error 0.1, the per-slice level is $0.1/40 = 0.0025$. A zero-event slice therefore needs 131 observations to certify 0.045 and 117 to certify 0.05.

The terminal-reference false-negative check defines the full-evidence comparator. The policy-tail audit limits concentration of excess automated-negative harm induced by a routing policy.

References

[1] Abhineet Agarwal, Yan Shuo Tan, Omer Ronen, Chandan Singh, and Bin Yu. Hierarchical shrinkage: Improving the accuracy and interpretability of tree-based models. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.

[2] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.

[3] Metod Jazbec, Alexander Timans, Tin Hadži Veljković, Kaspar Sakmann, Dan Zhang, Christian A. Naesseth, and Eric Nalisnick. Fast yet safe: Early-exiting with risk control. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[4] Liran Ringel, Regev Cohen, Daniel Freedman, Michael Elad, and Yaniv Romano. Early time classification with accumulated accuracy gap control. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 42584–42600, 2024.

[5] Kirill Trapeznikov and Venkatesh Saligrama. Supervised sequential classification under budget constraints. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2013.

[6] Joseph Wang, Kirill Trapeznikov, and Venkatesh Saligrama. An LP for sequential learning under budgets. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2014.